

Data Mining

309AA



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Class, meet Data



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Data comes from diverse sources, and generally is not tailor-made for some downstream task. We need to start from basics:

- What features are available?
- What are they measuring, exactly?
- What properties do they have?
- What are their relations?
- Are there outliers?
- ...

Data, in all shapes and sizes



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Data can be of different nature:

- Temporal: the data describes events *over time*
- Sequential: the data spans some ordering
- Relational: the data describes event *in between instances*
- Spatial: the data describes space
- Independent: instances in data are independent observations

These can co-occur!

Shapes in our thought exercise



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You are given a cycling data collection, with data gathered from different sources, covering all tours of thousands of cyclists from 2018 to 2024.

Data

- Speed
- Cadence
- Bike used
- Track, e.g., length, elevation, climbs
- Info on the cyclist, e.g., age

Shapes

- Independent: each activity is independent of the others
- Temporal: speed at each point in a given activity
- Sequential: the sequence of climbs in an activity
- Relational: groups of cyclists riding together
- Spatial: the GPX data itself

A GPX file gathers all GPS points, tracing movements on a map.

Categorizing data collections



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We refer to single instances in the collections as *objects/records/instances*, which are described by attributes.

Id	Age	Income	Marital	Loan grant
0	30	2.5k	Married	Yes
1	24	1.4k	Single	No
...	...	...	...	...

Attributes: Id , Age , Income , Marital ,
Loan grant

Records: 0 , 30 , 2.5k , Married , Yes , 1 , 24 ,
1.4k , Single , No

Attributes are often also called 'features', 'variables' in other fields, e.g., machine learning or statistics.

Data understanding



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After we have the gathered data, before cleaning and processing it, we need to understand it.

We start with the data and feature types, and how they are often represented.

Data types: Tabular



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When records are independent, and described by the same finite set of features, they are often represented in a *tabular* form: the *data matrix*. Each row is a record, each dimension is an attribute.

Id	Age	Bike used	Length	Duration	Date	Cyclist
0	28	Colnago VRS4	152.4	3:43:12	15-5-2025	Alessandro Covi
1	40	Cervelo RS5	72.4	2:55:01	4-3-2024	Gianni Affino

Records on the rows, attributes on the columns.

Data types: Transaction

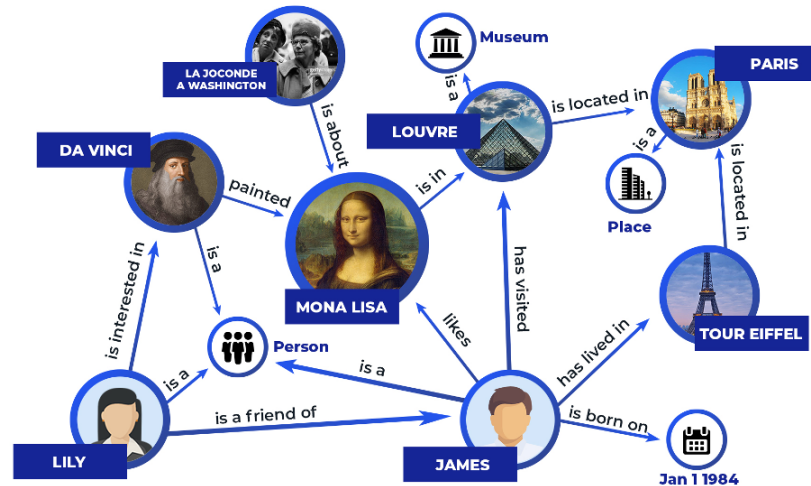
A feature contains a (multi)set of *items*.

Purchase Id	Cart	Bought on
0	Bread, Milk	17:12-15-5-2025
1	Notebook, Pens, Bread, Basil	8:04-4-3-2024

Records on the rows, attributes on the columns.

Data types: Graph

Data is linked, either on records or features.



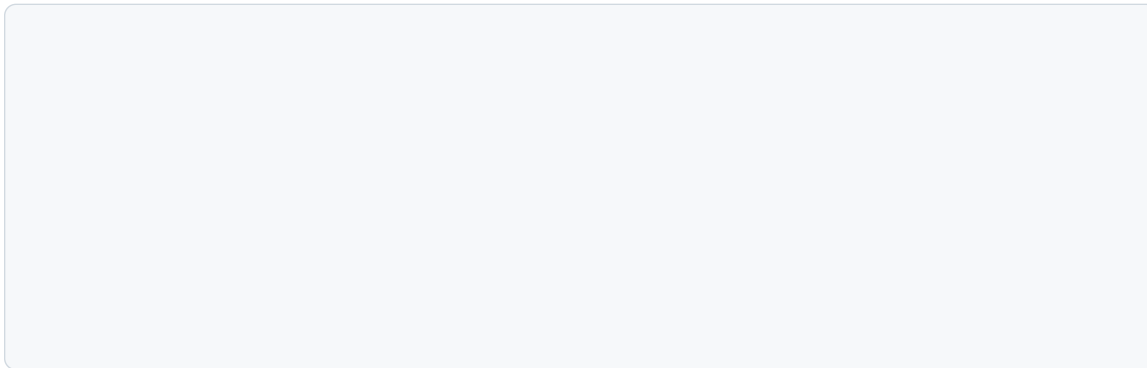
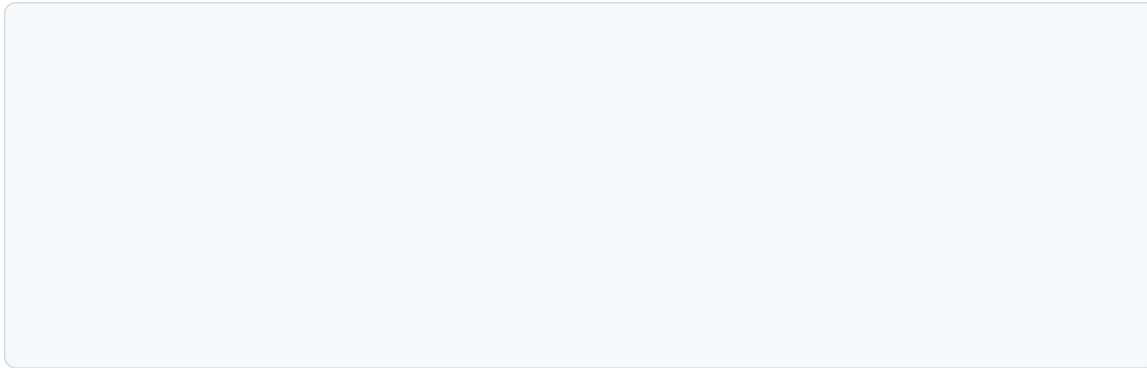
A simplified view of the Wikipedia knowledge graph on Mona Lisa.

Records are nodes in a graph, attributes can vary wildly across records.

Data types: Sequential

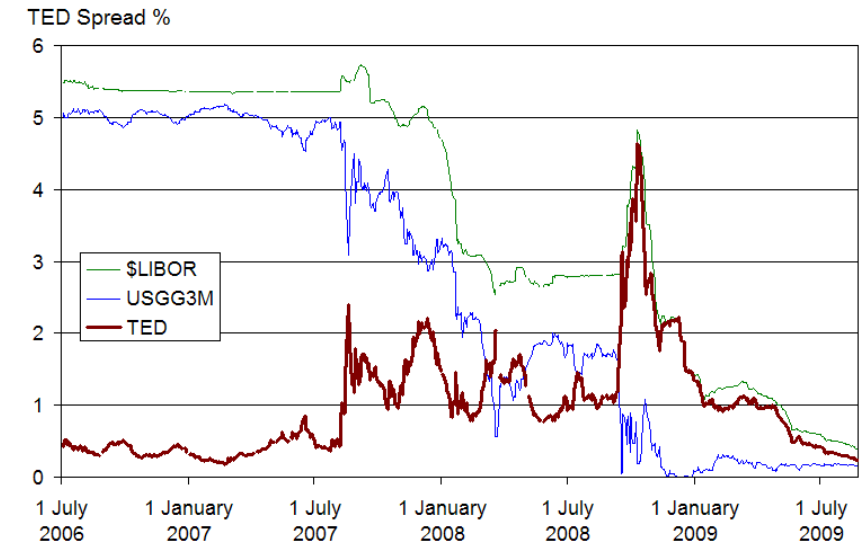
Records are sequences (of variable length): attributes are indexed (order or time).

Order



Start of the Divina Commedia (top) and codons synthesizing Glutamic acid (bottom).

Time

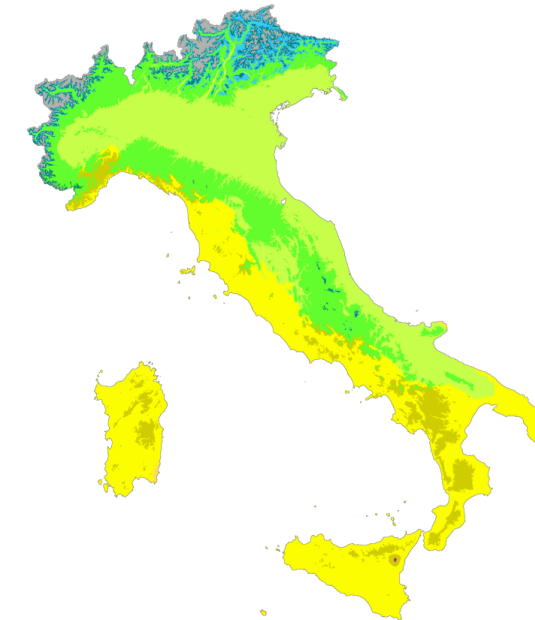


Risk in the financial system over time. Courtesy of Lawrence Khoo (Wikimedia).

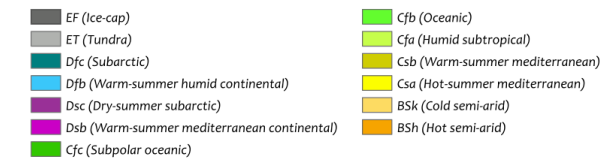
Data types: Spatial

Attributes are replaced by spatial indexing.

Köppen climate types of Italy



Köppen climate type



*Isotherm used to separate temperate (C) and continental (D) climates is -3°C
Data source: Climate types calculated from data from WorldClim.org

Climate types of Italy. Courtesy of Adam Peterson (Wikidata). CC BY-SA 4.0.

Attributes have types too!



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Type	Description	Example
Numerical	Values have a total ordering, and represent some numerical quantity	Age, dates
Ordinal	Values have a total ordering, and represent some quantity	Dress size, Cup size
Binary	Values are one of two categories: no ordering	Boolean values
Categorical	Values of one of multiple categories: no ordering	Country, Job

Attributes have types too!



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Type	Operations	Example
Numerical	Standard mathematical operators and functions	Mean, Max, +
Ordinal	Standard mathematical operators and functions, when appropriate	Max
Binary	Equality operators	=, ≠
Categorical	Equality operators	=, ≠

Values have types too!



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Values are either:

Discrete

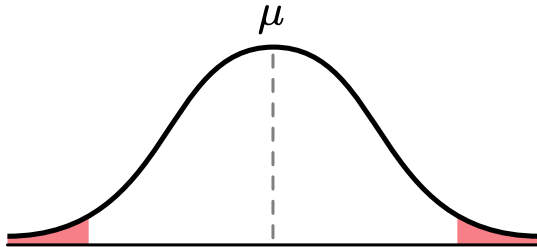
Defined in a finite or countably finite domain, e.g., country, job, cup size. Note: ordinal values may be discrete too!

Continuous

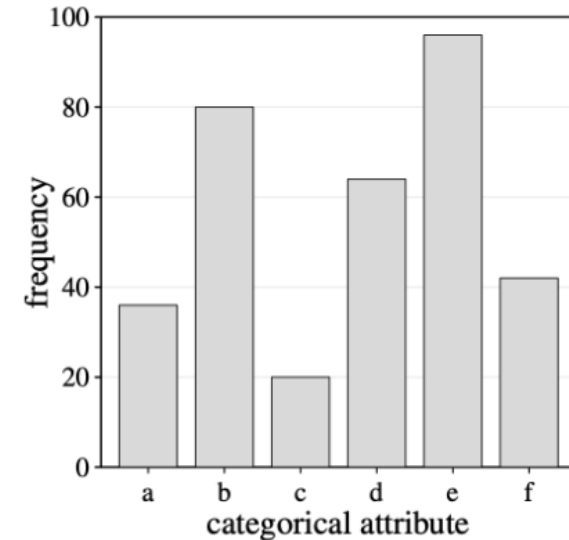
Defined in a continuous and infinite domain, e.g., distance.

Data syntax and semantics

Given the categorization of the records and attributes of your data, we can study its general behavior. We leverage some basic statistical tools, first of all by drawing the empirical distribution of the attributes.



Estimated distribution of a continuous attribute.



Distribution of a categorical attribute in a bar chart.

Data semantics: useful statistics



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Expected value

A statistic representative of the value of an attribute, weighing values and their probability

Variance

Distance from the expected value of all records: the data spread

Quantiles

Inflection points defining values for a threshold, e.g., if the 99-th percentile is 84, then we

Interquantile range

Distance between quantiles: how spread are inflection points?

Data semantics: useful statistics



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Expected value

$$\mathbb{E}[X] = \sum_{x \in \text{dom}(X)} \Pr(X = x)x$$

Variance

$$\sigma^2(X) = \mathbb{E}[\sum_{x \in \text{dom}(X)} (x - \mathbb{E}[X])^2]$$

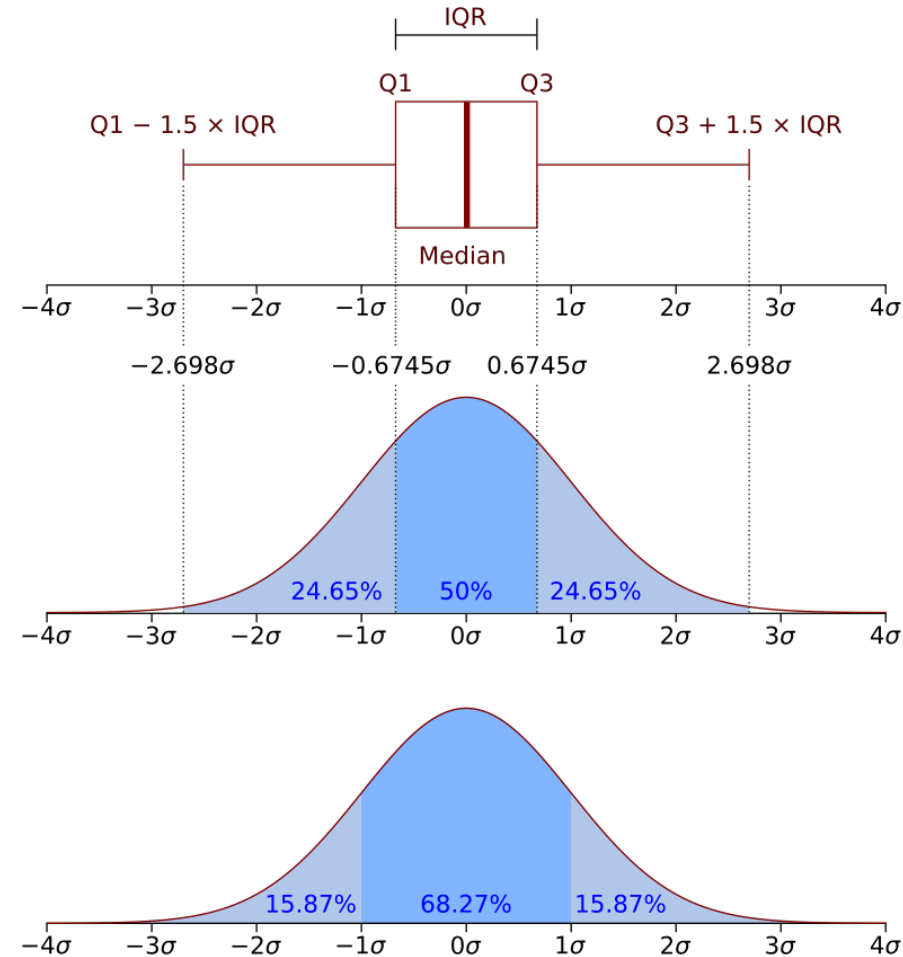
Quantile

$$q^p = x \text{ s.t. } \Pr(X \leq x) = q^p$$

Interquartile range

$$q^{75} - q^{25}$$

Data semantics: useful statistics



A Normal distribution, its q^{25} and q^{75} quantiles, and interquartile range between them. On top, a box plot visualization of the distribution.

Data semantics: closing tips



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- Statistical summary of the distribution are typically accompanied by visual and semantic one
- Erroneous or weird values, to be cleaned later can already pop up in these basic steps
- Outlier values typically skew statistics. Variance is often replaced by absolute/median average deviation

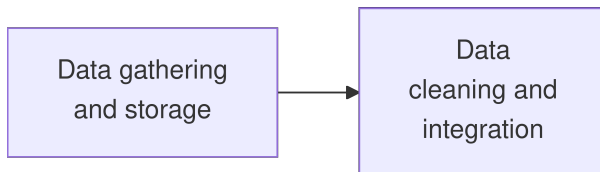
Data understanding in the KDD loop



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Determine the...

1. ... shape of your data
2. ... attributes of your data
3. ... types of your attributes
4. ... semantics of your attributes



The knowledge discovery pipeline, step 1.

We need to **clean** our data!

Data cleaning: sources of problems



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Data accuracy

- Syntactic: values outside domain, e.g., Eataly in Country
- Semantic: values in domain, but semantically wrong, e.g., age is 3, and weight is 82kg

Completeness

Some attributes are not collected, or are collected partially, e.g., temperature was not recorded by the sensor.

Biased gathering

Records may over/under-representative, e.g., the bank may only provide data about successful loan applicants.

Timeliness

Data is not up to date.

Data cleaning



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Remember: **garbage in, garbage out!** In a task-agnostic view, we are interested in addressing the above by tackling:

- Duplicates: skews the data distribution
- Missing values: give false/partial information
- Noise: uninformative of the data
- Poor accuracy: gives wrong data
- Outliers: skews the data distribution and models of the data

Dealing with... duplicates



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Trivial: remove them... when appropriate! Not all duplicates are garbage, it depends on what insight you can gather from it.

Case A

You have data on registration to your website, with several duplicate e-mails.

Insights:

- The "Sign in" button is hard to find
- The "Sign in" button is less visible than the "Sign up" button
- Your site is so anonymous people forget they signed up already

Case B

You have data on credit account opening from Poste (Italian postal service) with several duplicate e-mails.

Insights:

- The client hacked the database and added themselves to ask more credit (unlikely)
- Poste's tech staff is underwhelming (very likely)

Dealing with... duplicate features?

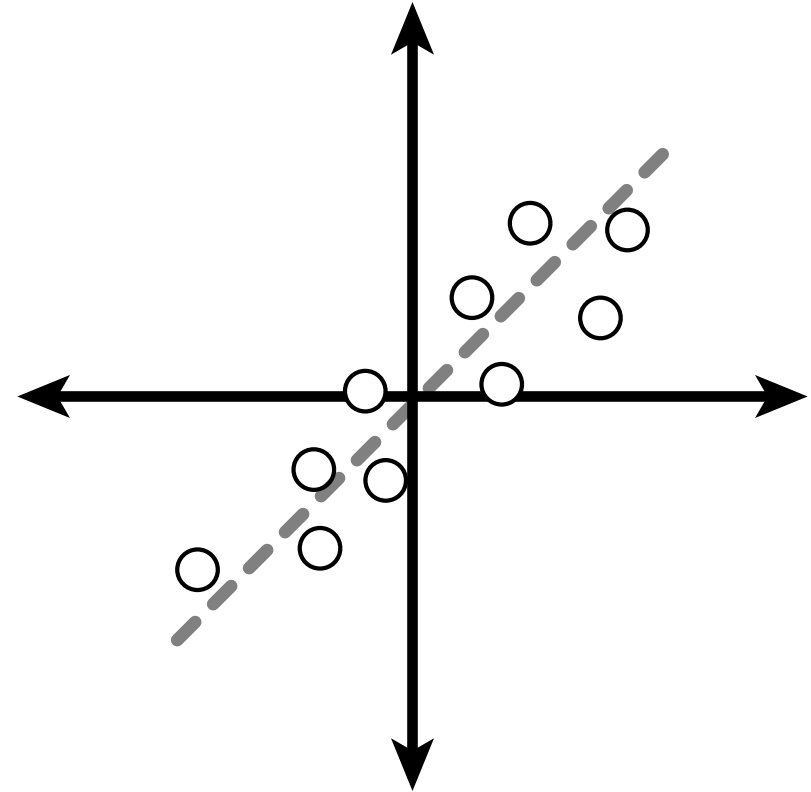


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Features convey similar, although not equal, information to others.

- Resting heart rate and heart rate under continuous high effort
- Education level and reading skills
- Rent and available bank deposit

These pairs of features are not per se one duplicate of the other, but are strongly related: when one grows, so does the other, and when one goes down, so does the other.



Correlation between two features: as one grows, so does the other.

Dealing with... duplicates?



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Linear (and rank) relationships between two features X, Y can be quantified with their correlation. Correlation ranges in $[-1, +1]$, from perfectly negative to perfectly positive correlation.

Given two lists of values x^{i^n}, y^{i^n} , we can compute two main correlation types.

Pearson

Purely numerical, applicable to numeric features.

$$\rho_P^{X,Y} = \frac{\mathbb{E}[(x^i - \mathbb{E}[X])(y^i - \mathbb{E}[Y])]}{\sigma_X \sigma_Y}$$

Spearman

Pearson, applied to the *rank* of feature values, i.e., their relative position within their values.

$$\rho_S = \rho_P^{\text{rank}(X), \text{rank}(Y)}$$

Dealing with... missing values



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Data may be missing for any number of reasons (at random or not at random).

1. A record has a large and/or significant set of missing attributes
2. An attribute has a large percentage of missing values

We have two choices: dropping or imputing.

Dealing with... missing values



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Dropping

- High percentage of missing values
- Missing values in critical attributes, e.g., a patient in cardiology has no heart rate data

vs

Imputing

- Low percentage of missing values
- Reasonably good understanding of the attribute semantics/distribution
- Presence of related attributes

We create a model to *predict* the missing value

Dealing with... outliers

Quantiles and distributions inform us on what values may be outlier. They are typically dropped, and unlike missing values, almost never imputed.

We'll tackle algorithms later in the course.

From number to pictures: visualization



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Yet another example: Iris dataset detailing the sepal length and widths, and petal length and width of 150 Iris Setosa, Iris Versicolor, Iris Virginica.

Sepal L.	Sepal W.	Petal L.	Petal W.	Type
5.1	3.5	1.4	0.2	Setosa
7.0	3.2	4.7	1.4	Versicolor
...	...	...	...	...



iris setosa



iris versicolor

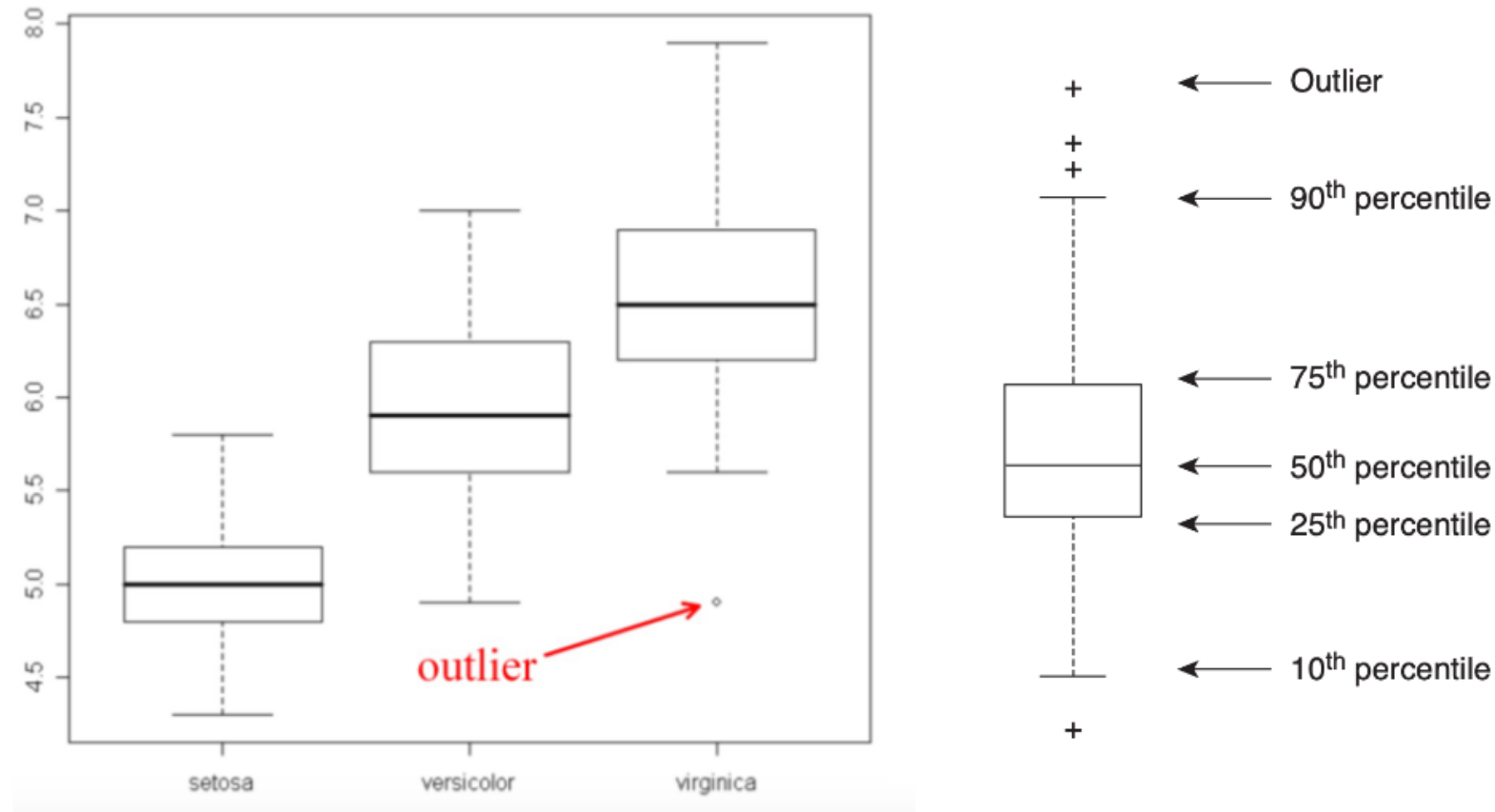


iris virginica

The three Iris types in the dataset.

Box plot

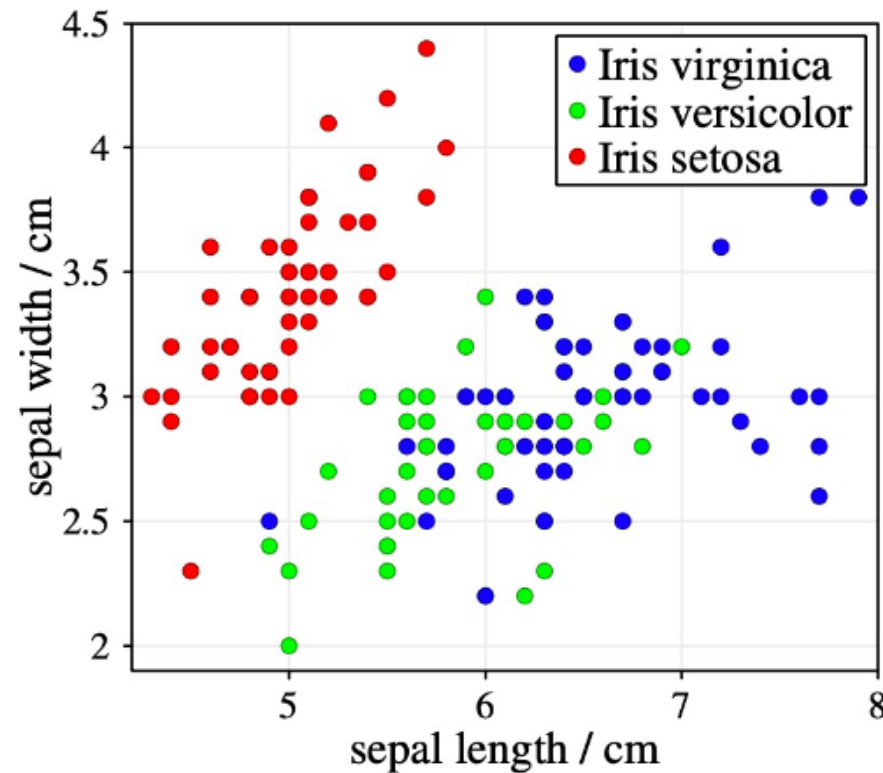
Plot univariate data, eyeing outliers.



A box plot of the sepal length in the Iris dataset. The bold bar indicates the mean value, the box q^{25} and q^{75} , the bars q^{10} and q^{90} . Remaining instances are represented as circles.

Scatter plot

Plot bivariate (or trivariate) data, eyeing data correlation and outliers.



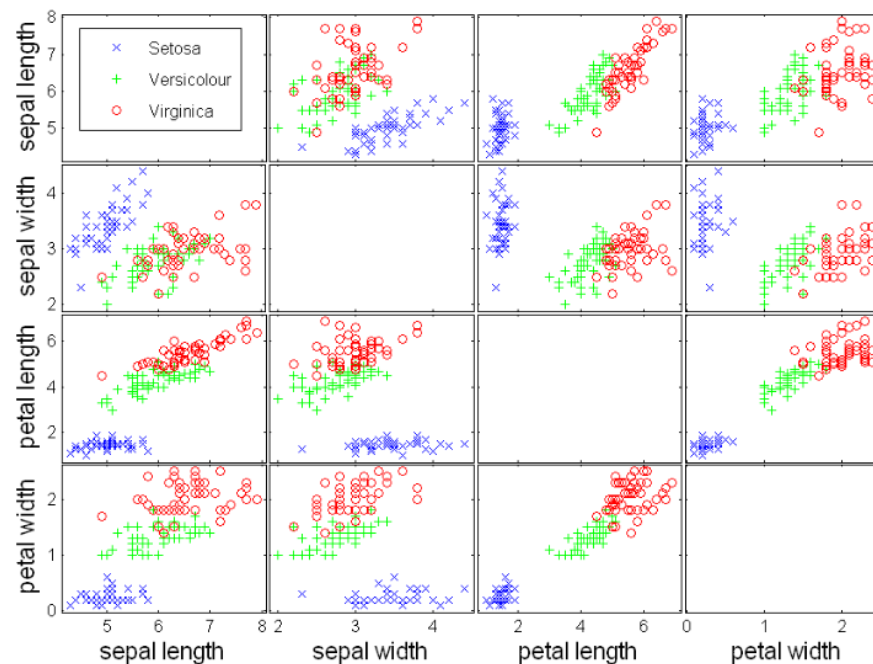
A scatter plot of the sepal length and width in the Iris dataset.

Scatter plot



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Plot bivariate (or trivariate) data, eyeing data correlation and outliers.



A scatter matrix: scatter plots of all pairs of attributes in the Iris dataset.